ZCAS UNIVERSITY

SCHOOL OF COMPUTING, TECHNOLOGY & APPLIED SCIENCES

CAS6021 DIGITAL ANALYTICS: WEBSITE DESIGN AND GOOGLE ANALYTICS

FINAL EXAMINATION

WEDNESDAY, 27 NOVEMBER, 2024

09:00 – 12:00 HOURS

TIME ALLOWED: WRITING – THREE HOURS: READING – 5 MINUTES

INSTRUCTIONS:

1. Section A: This question is **compulsory** and must be attempted.
2. Sections B: Answer **TWO (2)** questions from this section.
3. This examination paper carries a total of **100 marks**.
4. Candidates must **not turn this page** until the invigilator tells them to do so.

Decision

SECTION A: Question 1 is compulsory and must be attempted

Question 1

Read the following case scenario and answer the questions below.

Scenario:

A fashion e-commerce website specialising in trendy, affordable apparel has been investing heavily in social media marketing, leading to a steady increase in visitor traffic. This growth has been particularly driven by visually appealing ads and influencer partnerships on platforms like Instagram and TikTok. Despite the traffic increase, the website's analytics reveal a significant issue: visitors are not engaging deeply with the product catalog. Most users land on a category page (e.g., "Summer Dresses" or "Men's Casual Wear") but then leave without exploring specific product pages. On average, they view only one or two products, resulting in a high bounce rate on the category pages and a low overall conversion rate.

To improve engagement, the company aims to increase the click-through rate to individual product pages, reduce the bounce rate on category pages, and ultimately raise the conversion rate from casual visitors to purchasers.

a) Identify and explain three possible reasons why visitors may not be exploring products beyond the initial category pages. **(18 Marks)**

b) Suggest two changes to the category pages that could improve user engagement and encourage deeper browsing. Briefly explain each suggestion and how it may influence user behaviour. **(12 Marks)**

c) Explain the relevance of each of the following metrics in analysing user engagement on category pages:

   i. Click-Through Rate to Product Pages **(5 Marks)**

   ii. Pages per Session **(5 Marks)**

   iii. Bounce Rate **(5 Marks)**

d) Describe one advantage of using session recordings to understand user interactions on category pages. **(5 Marks)**

[Total: 50 Marks]

SECTION B: Attempt any TWO (2) questions in this section

Question 2

a) Define "***user flow***" and explain why it is critical in designing an effective e-commerce website. **(5 Marks)**

b) Discuss how each of the following elements contributes to creating an optimized user flow:

i. Simple Navigation **(5 Marks)**

ii. Product Page Layout **(5 Marks)**

c) Provide three recommendations for improving user flow on an e-commerce website, and briefly explain how each recommendation could enhance the user experience and support conversions. **(10 Marks)**

[Total 25 Marks]

Question 3

a) Explain the concept of "***data visualization***" and its importance for interpreting analytics data in digital marketing. **(5 Marks)**

b) Describe how each of the following types of data visualizations could help a digital marketing team make informed decisions:

i. Funnel Visualization **(5 Marks)**

ii. Heatmaps **(5 Marks)**

c) Provide two recommendations for effectively using data visualizations to present analytics insights to stakeholders, and briefly explain the value of each recommendation.

(10 Marks)

[Total: 25 Marks]

Question 4

a) Define "***customer segmentation***" and discuss its importance for delivering personalized marketing. **(5 Marks)**

b) Explain how each of the following segmentation criteria can enhance targeting efforts in a digital marketing campaign:

i. Behavioural Segmentation **(5 Marks)**

ii. Demographic Segmentation **(5 Marks)**

c) Propose two ways customer segmentation data can be used to create personalized marketing messages, and briefly explain the potential impact of each approach on customer engagement. **(10 Marks)**

[Total: 25 Marks]

END OF EXAMINATION